



# LNG Cargo Diversion Decision Engine

Structured decision framework for Europe vs Asia LNG diversion economics.

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# 1

## Executive Summary

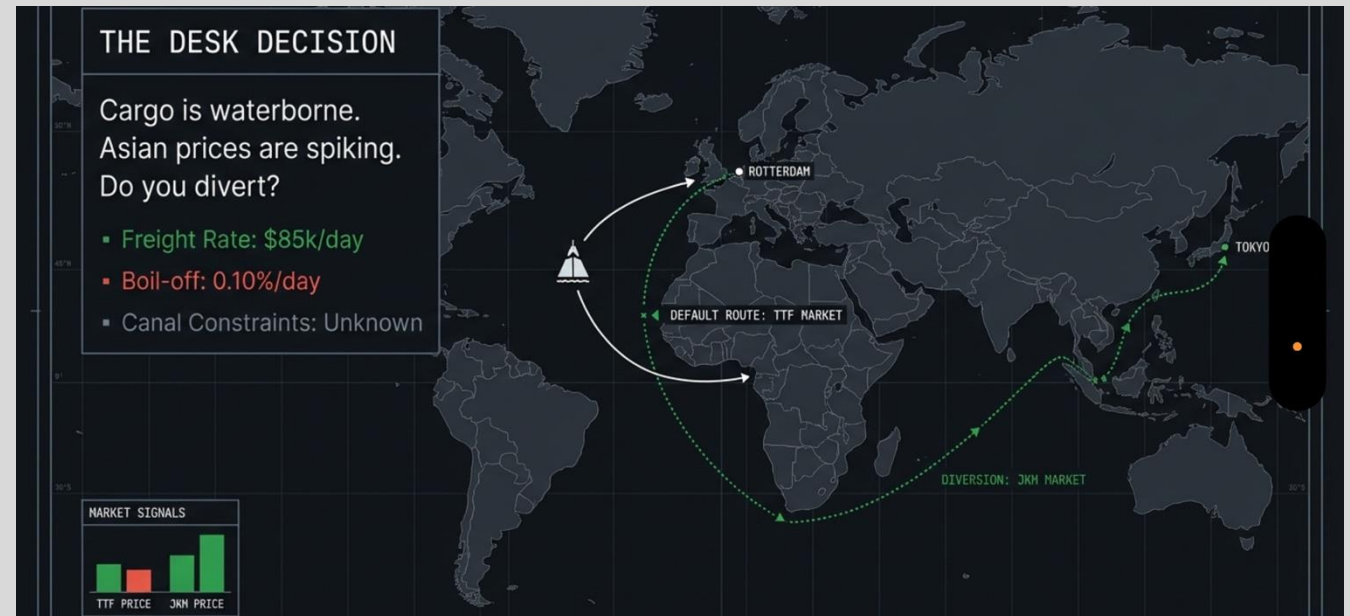
# Problem

Given an LNG cargo loading in the US Gulf

Should the vessel discharge in Europe (TTF) or divert to Asia (JKM)?

Decision requires integrating:

- Price spreads
- Freight economics
- Carbon costs
- Operational risk buffers
- Hedge execution considerations



# Approach

A rule-based decision engine that:

- Calculates netback economics
- Applies conservative risk adjustments
- Generates executable hedge instructions
- Validates behavior via historical rule testing

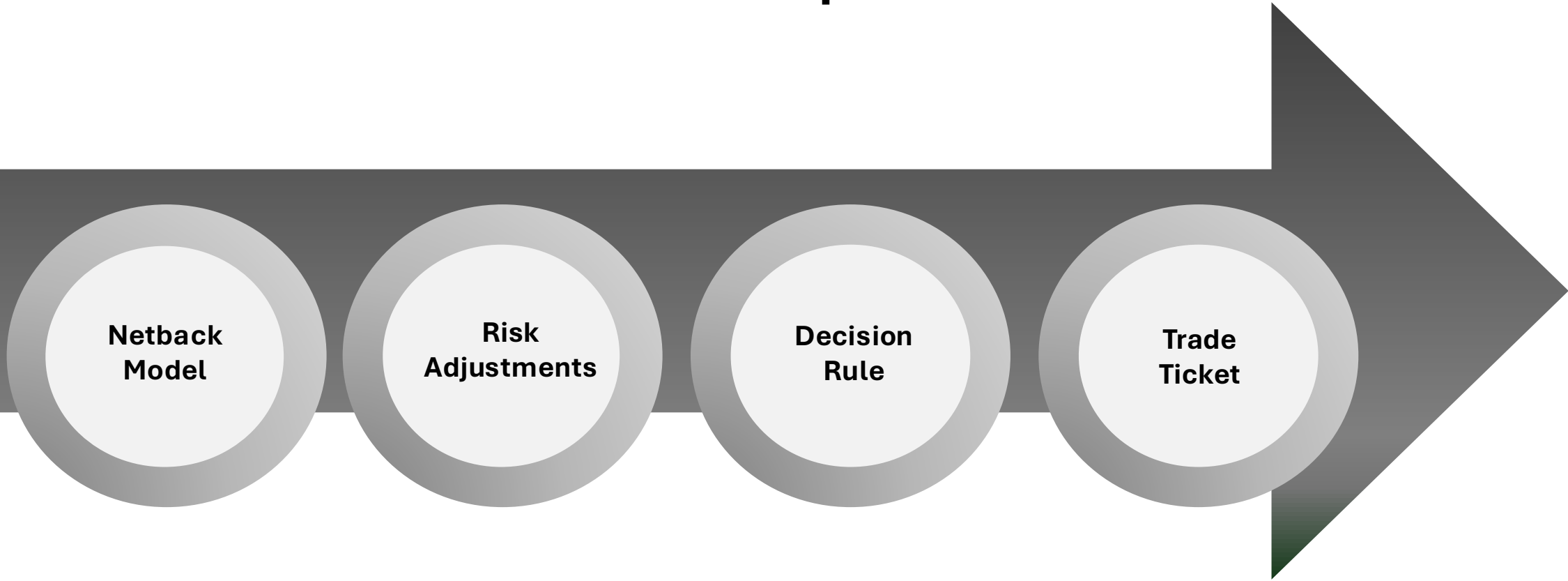
# Example Output

| DIVERT TRIGGERED |                                      |
|------------------|--------------------------------------|
| Europe Netback   | \$126.26 M                           |
| Asia Netback     | \$133.96 M                           |
| Adjusted Uplift  | \$7.07 M                             |
| Hedge            | BUY JKM 322 lots / SELL TTF 322 lots |

# 2

## Decision Framework

# Decision Framework - Market Inputs



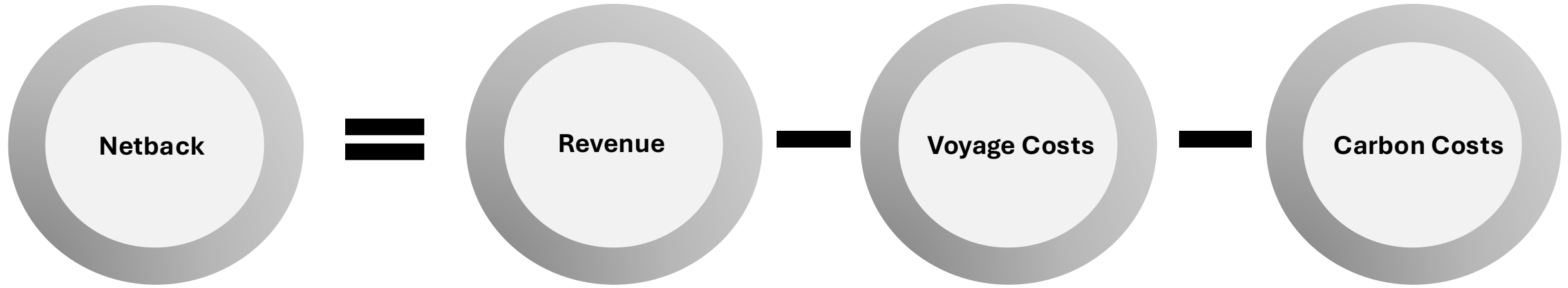
- Raw uplift = Asia netback – Europe netback
- Basis haircut applied
- Operational buffer deducted
- Decision triggered only above threshold.



# 3

## Netback Methodology

# Netback Calculation Logic



- Distance ➡ Voyage time
- Revenue = Destination Price x Delivered LNG Energy
- Fuel consumption
- Boil-off adjustment
- EUA exposure
- Delivered energy

# 4

## Example Diversion Decision

# Example Trade Output

1. Clear decision
2. Hedge sizing
3. Adjusted uplift logic

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## LNG DIVERSION TRADE NOTE

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Route: US\_Gulf → Rotterdam (TTF) vs Tokyo (JKM)  
Vessel: TFDE | Cargo: 174,000 m<sup>3</sup>  
As of: 2026-02-23 00:31 UTC

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TTF: \$32.03/MMBtu (real)  
JKM: \$34.78/MMBtu (proxy)  
Freight: \$85,000/day (proxy)  
Fuel: \$747/t (proxy)  
EUA: \$69.21/tCO<sub>2</sub> (real)

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Europe netback: \$126,771,038  
Asia netback: \$134,466,238  
Raw uplift: \$7,695,200  
Adjusted uplift: \$7,060,440  
(basis=5.0%, ops=\$250,000, threshold=\$500,000)

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✓ Decision: DIVERT

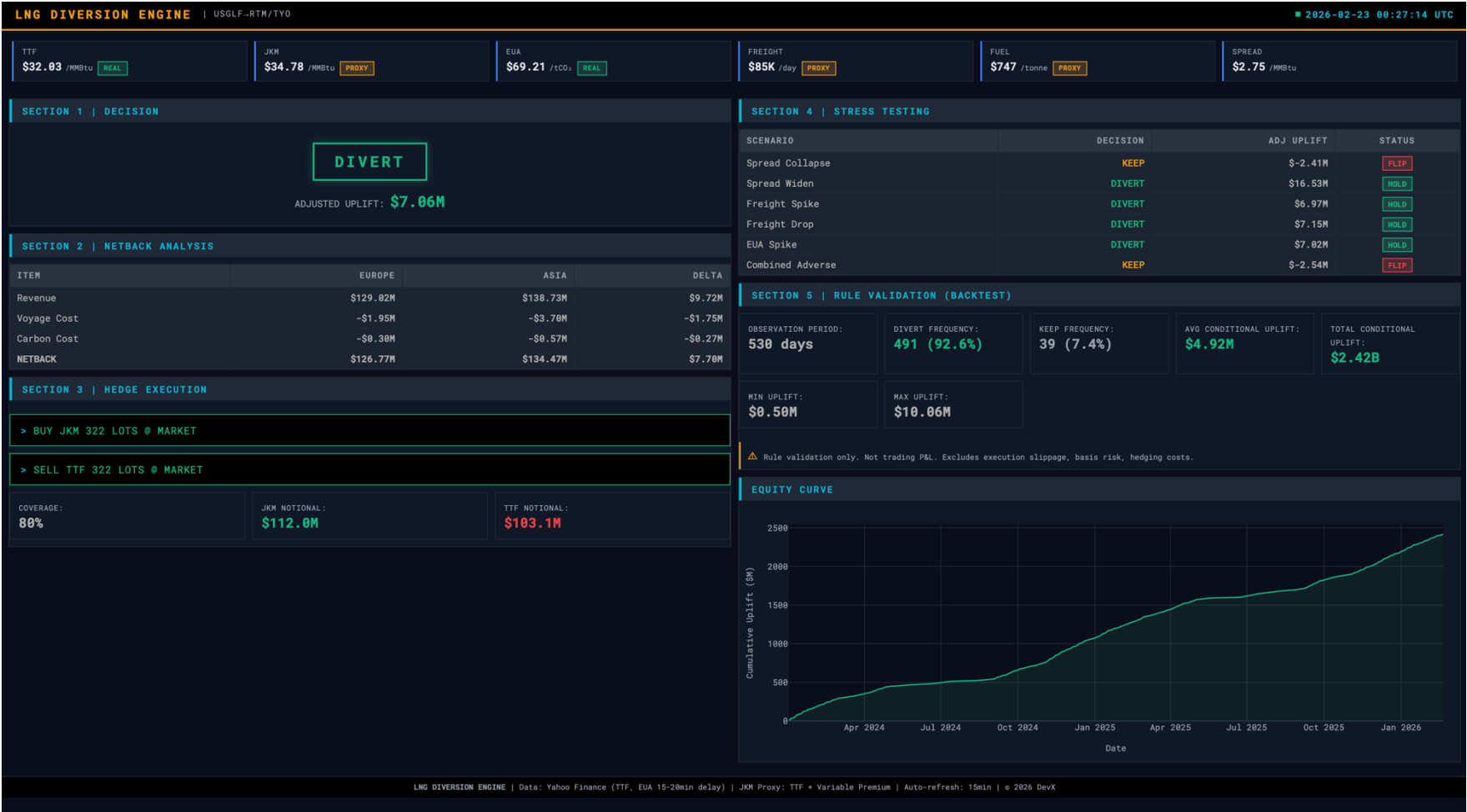
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Hedge: BUY JKM 322 lots | SELL TTF 322 lots

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# Terminal Dashboard - LNG Diversion Engine

- 1. Route: US Gulf to Rotterdam vs Tokyo
- 2. Adjusted uplift: \$7.06M
- 3. Decision: DIVERT
- 4. Hedge: BUY JKM / SELL TTF



# 5

## Risk & Stress Testing

# Decision Robustness

| SCENARIO         | DECISION | ADJ UPLIFT | STATUS |
|------------------|----------|------------|--------|
| Spread Collapse  | KEEP     | \$-2.41M   | FLIP   |
| Spread Widen     | DIVERT   | \$16.53M   | HOLD   |
| Freight Spike    | DIVERT   | \$6.97M    | HOLD   |
| Freight Drop     | DIVERT   | \$7.15M    | HOLD   |
| EUA Spike        | DIVERT   | \$7.02M    | HOLD   |
| Combined Adverse | KEEP     | \$-2.54M   | FLIP   |

- Decision flips under spread collapse
- Demonstrates sensitivity to freight/carbon shocks

# 6

## Rule Behaviour (Historical Validation)



# Rule Behaviour Over Time

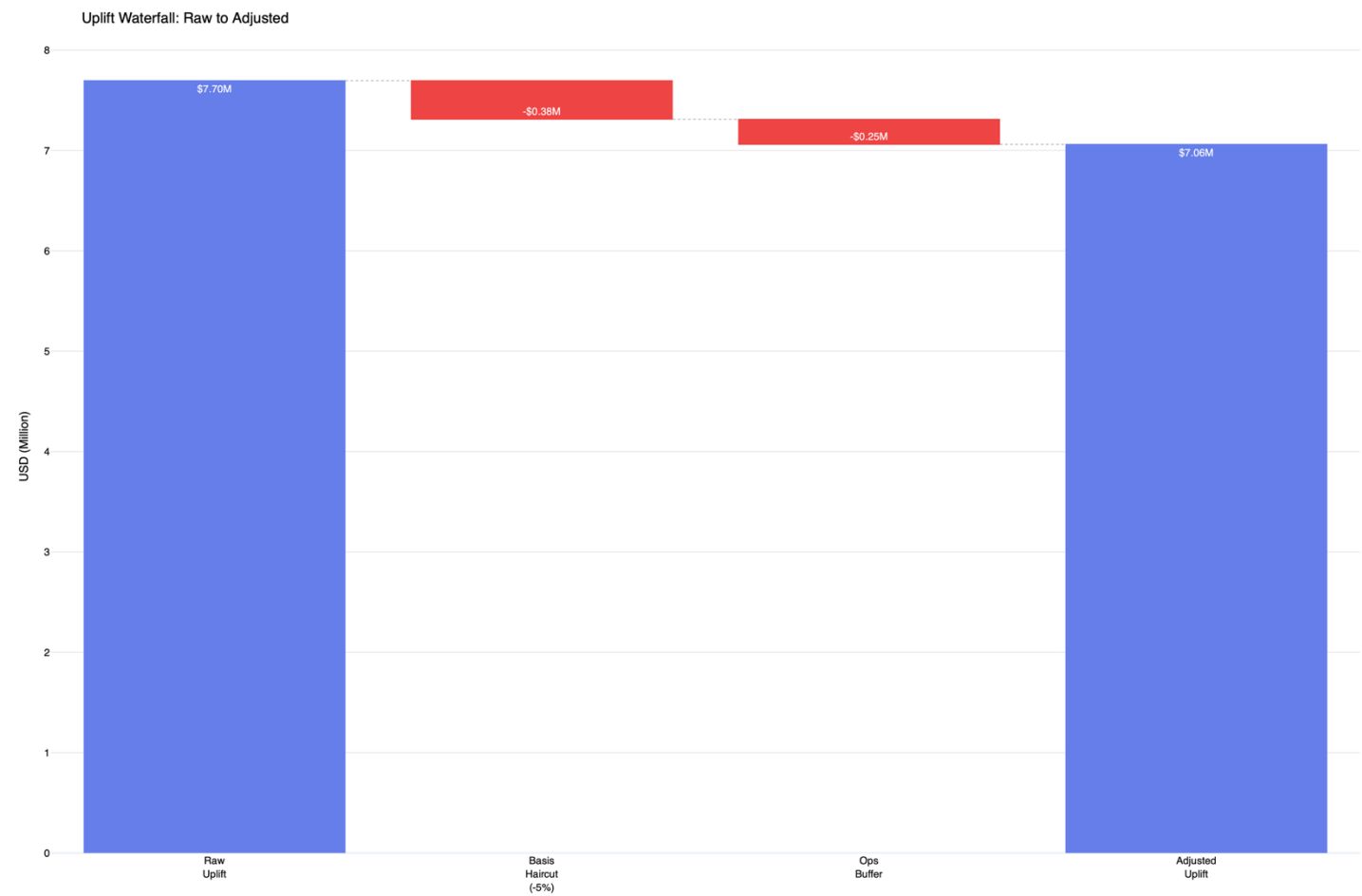


*Disclaimer: Historical simulation indicates diversion conditions frequently satisfied under proxy spread dynamics*

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## Decision Waterfall

# From Raw Spread to Executable Diversion Decision



# 8

## Assumptions & Next Steps

# Model Limitations and Future Enhancements

| Limitations                                  | Next Evolution                 |
|----------------------------------------------|--------------------------------|
| JKM Proxy                                    | Licensed JKM feed              |
| Freight Proxy                                | Freight curve integration      |
| Simplified Routing                           | Stochastic ops delays          |
| No demurrage modelling                       | Optionality modelling          |
| Contract structure (FOB vs DES) not modelled | Contract Aware diversion logic |